

Appendix F: BERT Selected Configurations

A SST-2 Final Architectures

The table reports two selected BERT-base encoder configurations found for SST-2. Each row is one encoder layer. Active heads are reported out of 12 attention heads, and active feed-forward neurons are reported out of 3072 intermediate neurons.

Reading guide. Darker cells retain more structure. The 3.7M-parameter model reaches 91.60 accuracy, while the 4.0M-parameter model reaches 92.43 accuracy.

L	3.7M, 91.60		4.0M, 92.43	
	Heads	Neurons	Heads	Neurons
0	4/12	112/3072	1/12	85/3072
1	2/12	144/3072	1/12	89/3072
2	2/12	64/3072	1/12	89/3072
3	1/12	112/3072	1/12	89/3072
4	0/12	96/3072	1/12	89/3072
5	0/12	48/3072	1/12	89/3072
6	0/12	64/3072	1/12	89/3072
7	1/12	16/3072	1/12	89/3072
8	1/12	32/3072	1/12	89/3072
9	0/12	48/3072	1/12	89/3072
10	0/12	80/3072	1/12	89/3072
11	1/12	64/3072	1/12	89/3072

B QNLI and MNLI Final Architectures

The table reports the selected tiny encoder configurations for QNLI and MNLI. The same layer-wise convention is used: active heads are reported out of 12, and active feed-forward neurons are reported out of 3072.

L	QNLI 3.2M, 88.20		MNLI 3.1M, 81.36	
	Heads	Neurons	Heads	Neurons
0	0/12	0/3072	2/12	112/3072
1	1/12	32/3072	3/12	96/3072
2	1/12	64/3072	2/12	64/3072
3	1/12	48/3072	2/12	64/3072
4	3/12	48/3072	0/12	16/3072
5	1/12	64/3072	0/12	16/3072
6	1/12	48/3072	0/12	0/3072
7	1/12	48/3072	1/12	32/3072
8	0/12	0/3072	0/12	32/3072
9	1/12	48/3072	1/12	32/3072
10	0/12	32/3072	0/12	16/3072
11	3/12	32/3072	1/12	16/3072

C Moderate QNLI and MNLI Architectures

The table reports the selected moderate-compression encoder configurations for QNLI and MNLI.

L	QNLI 34.1M, 91.90		MNLI 34.0M, 85.60	
	Heads	Neurons	Heads	Neurons
0	3/12	1408/3072	5/12	1168/3072
1	1/12	1552/3072	5/12	1056/3072
2	1/12	2800/3072	6/12	1408/3072
3	0/12	2128/3072	6/12	1504/3072
4	0/12	1504/3072	5/12	1504/3072
5	3/12	1760/3072	6/12	1280/3072
6	1/12	1152/3072	5/12	992/3072
7	3/12	2176/3072	6/12	1840/3072
8	2/12	1472/3072	6/12	1664/3072
9	0/12	1280/3072	6/12	752/3072
10	3/12	1520/3072	5/12	208/3072
11	0/12	1248/3072	5/12	192/3072

D Aggregate Structure

Configuration	Heads	Neurons	Zero-head layers
SST-2 3.7M, 91.60	12/144	880/36864	5
SST-2 4.0M, 92.43	12/144	1064/36864	0
QNLI 3.2M, 88.20	13/144	464/36864	3
MNLI 3.1M, 81.36	12/144	496/36864	5
QNLI 34.1M, 91.90	17/144	20000/36864	4
MNLI 34.0M, 85.60	66/144	13568/36864	0